Asset Pricing and Equity Premium Analysis

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Data source: FRBSF Quarterly TFP (1947:Q2 – 2026:Q1)

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Abstract

This report documents a computational macrofinance project developed during the course of Asset Pricing held by professor Martin Gonzalez Eiras at LMEC unibo master program.

Starting from raw U.S. TFP data, we (i) estimate an AR(1) process for TFP growth by OLS, (ii) discretize it into a 9 state Markov chain via the Rouwenhorst (1995) method, (iii) solve the resulting stochastic neoclassical growth model by value function iteration (VFI) for five values of the coefficient of relative risk aversion γ, and (iv) simulate 10,000 quarters of the model economy to compute asset pricing moments, the risk free rate, expected equity return, and equity premium. The model produces a near zero or slightly *negative* equity premium for all γ considered, illustrating the classic equity premium puzzle in a standard RBC framework.

All numerical results are reproducible from the accompanying Jupyter notebook `macrofinancepy.ipynb`.

1 Research question

Asset pricing in production economies requires a careful calibration of the exogenous technology process that drives fluctuations in output, consumption, and investment. The central question in this project is whether a standard RBC model with CRRA preferences can generate an empirically plausible equity premium. As Section 6 shows, it cannot, this is a manifestation of the Mehra and Prescott (1985) equity premium puzzle.

2 Data and TFP Growth Transformation

2.1 Source

The raw data are downloaded from the Federal Reserve Bank of San Francisco (FRBSF) TFP database, stored in `quarterly_tfp.xlsx` (sheet: `quarterly`).

2.2 Transformation to Quarterly Log Differences

The variable `dtfp` is an annualized percentage change. To obtain quarterly log differences suitable for log linearized models, we apply:

$$g_t = \frac{\texttt{dtfp}_t}{400}. \tag{1}$$

Observations with missing g_t are dropped, yielding the clean sample described in Table 1.

Table 1: Data Summary

Attribute	**Value**
Sample start	1947:Q2
Sample end	2026:Q1
Frequency	Quarterly
Usable observations (T)	316
Transformation	$g_t = \texttt{dtfp}_t/400$

3 AR(1) Estimation

3.1 Model

TFP growth is assumed to follow a stationary first order autoregression:

$$g_t \;=\; a + \rho\, g_{t-1} + \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0,\, \sigma_\varepsilon^2\right), \tag{2}$$

with stationarity requiring $|\rho| < 1$. Under stationarity, the unconditional moments are:

$$\mu_g = \frac{a}{1-\rho}, \tag{3}$$

$$\sigma_g^2 = \frac{\sigma_\varepsilon^2}{1-\rho^2}. \tag{4}$$

3.2 Estimation Method

The OLS regression is run on the $T-1 = 315$ pairs $\{(g_{t-1},\, g_t)\}_{t=2}^{316}$, using `statsmodels.OLS` with a constant. The innovation standard deviation is estimated as $\hat{\sigma}_\varepsilon = \sqrt{\widehat{\text{MSE}}}$, where MSE uses the $T-2 = 314$ degrees of freedom correction.

3.3 Results

Table 2: OLS Estimates of the AR(1) for TFP Growth

Parameter	**Symbol**	**Estimate**	**Annualized equiv.**
Intercept	a	0.002590	—
Persistence	ρ	0.144172	≈ 0.0004[a]
Innovation std. dev.	σ_ε	0.008882	$\approx 3.55\%$
Implied unconditional moments			
Unconditional mean	μ_g	0.003027	$\approx 1.21\%$
Unconditional std. dev.	σ_g	0.009024	$\approx 3.61\%$

Intercept ($a = 0.00259$). Implies a long run quarterly TFP growth rate of $\mu_g \approx 0.30\%$ (1.21% annualized), broadly consistent with post war U.S. growth accounting.

Persistence ($\rho = 0.144$). Shocks are largely transitory at the quarterly frequency. This low persistence is typical of quarterly TFP estimates and reflects measurement noise; annual RBC calibrations routinely use $\rho \approx 0.95$ on annual data.

Volatility ($\sigma_\varepsilon = 0.00888$). The implied annualized TFP volatility of $\approx 3.6\%$ is consistent with the Cooley and Prescott (1995) calibration range.

4 Rouwenhorst Discretization

4.1 Why Rouwenhorst?

Tauchen (1986) works well for low persistence processes but deteriorates as $\rho \to 1$. Kopecky and Suen (2010) show that the Rouwenhorst (1995) method matches the first two unconditional moments *exactly* for any ρ, and is therefore preferred in quantitative work.

4.2 Algorithm

Given N states, define:

$$\sigma_z = \frac{\sigma_\varepsilon}{\sqrt{1-\rho^2}}, \qquad \psi = \sqrt{N-1}\,\sigma_z, \qquad p = q = \frac{1+\rho}{2}. \tag{5}$$

State grid. Evenly spaced around the unconditional mean:

$$z_i = \mu_g - \psi + \frac{2\psi(i-1)}{N-1}, \quad i = 1, \ldots, N. \tag{6}$$

4.3 State Grid ($N = 9$)

Table 3: Rouwenhorst State Grid for TFP Growth ($N = 9$)

Index i	**State** z_i	**Economic interpretation**
1	-0.022361	Severe contraction
2	-0.016014	Strong contraction
3	-0.009667	Mild contraction
4	-0.003320	Slight contraction
5	0.003027	Long run mean (μ_g)
6	0.009374	Slight expansion
7	0.015721	Mild expansion
8	0.022068	Strong expansion
9	0.028414	Boom

Uniform grid spacing $\Delta z = \psi/(N-1) \times 2 \approx 0.006347$. State 5 equals the unconditional mean $\mu_g = a/(1-\rho) = 0.003027$.

4.4 Moment Validation

The stationary distribution $\boldsymbol{\pi}$ (computed via $\mathbf{P}^{1000}$) is used to verify that the discrete chain replicates the continuous AR(1) moments exactly.

Table 4: Continuous vs. Discrete Moment Comparison

Moment	**Continuous AR(1)**	**Rouwenhorst chain**
Mean μ_g	0.00302685	0.00302685
Variance σ_g^2	0.00008057	0.00008057

Exact match to 8 decimal places confirms the Rouwenhorst property. Continuous formulas: $\mu_g = a/(1-\rho)$, $\sigma_g^2 = \sigma_\varepsilon^2/(1-\rho^2)$.

5 Value Function Iteration

5.1 Model

The representative household maximises expected discounted utility from consumption:

$$\max_{\{c_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad u(c) = \frac{c^{1-\gamma} - 1}{1-\gamma}, \tag{7}$$

subject to the resource constraint:

$$c_t + k_{t+1} \;=\; Z_t\, k_t^{\alpha} + (1-\delta)\, k_t, \qquad Z_t = e^{z_t}, \tag{8}$$

where z_t follows the Markov chain described in Section 4.

5.2 Calibration

Table 5: Structural Parameters

Parameter	**Description**	**Value**	**Freq.**
α	Capital share	0.33	—
β	Discount factor	0.99	Quarterly
δ	Depreciation rate	0.025	Quarterly
γ	CRRA coefficient	{0.5, 1.5, 3.0, 5.0, 10.0}	—
a	AR(1) intercept	0.002590	Quarterly
ρ	AR(1) persistence	0.144172	Quarterly
σ_ε	AR(1) shock s.d.	0.008882	Quarterly

5.3 Numerical Setup

The deterministic steady state satisfies:

$$k_{ss} = \left(\frac{\alpha}{1/\beta - (1-\delta)} \right)^{1/(1-\alpha)}. \tag{9}$$

The capital grid is $k \in [0.5\, k_{ss},\ 1.5\, k_{ss}]$ with $n_k = 500$ equally spaced points. The Bellman equation is solved by exhaustive grid search on the full $n_k \times N_z \times n_k$ tensor of continuation values. Expected values are computed as $EV = V\,\mathbf{P}^{\top}$.

The Bellman equation at each grid point (k, z) is:

$$V(k,z) = \max_{k' \in \mathbf{k}} \left\{ u(Zk^{\alpha} + (1-\delta)k - k') + \beta\, \mathbb{E}[V(k', z') \mid z] \right\}. \tag{10}$$

Infeasible choices ($c \leq 0$) are penalised with $U = -10^{10}$.

5.4 Convergence

Iterations stop when $\|V_{\text{new}} - V\|_\infty < 10^{-6}$. Table 6 reports convergence iterations for each γ.

Table 6: VFI Convergence Iterations

γ	**Iterations to convergence**
0.5	1,371
1.5	1,332
3.0	1,283
5.0	1,233
10.0	1,156

Tolerance 10^{-6}; max iterations 1,500. Higher γ converges faster because the value function is more concave and the policy is more interior.

5.5 Consumption Policy Function

The consumption policy $c(k, z)$ is plotted in the notebook for $\gamma \in \{0.5, 10.0\}$ and three TFP states (low, neutral, high). Key qualitative features:

- $c(k, z)$ is *increasing* in both k and z, richer agents and better productivity both raise consumption.
- The policy is steeper under $\gamma = 0.5$ (risk tolerant agent) than under $\gamma = 10.0$ (risk averse agent), reflecting a larger marginal propensity to consume when the precautionary motive is weak.
- The vertical gap between TFP states widens at higher capital levels, since capital amplifies the productivity differential.

6 Simulation and Asset Pricing Moments

6.1 Simulation Design

The model is simulated for $T = 11{,}000$ quarters; the first 1,000 are discarded as burn in, leaving $N_{\text{sim}} = 10{,}000$ usable observations. The TFP path $\{z_t\}$ is drawn from the Markov chain starting at the middle state, using `np.random.seed(323)` for reproducibility. Capital and consumption paths follow the VFI policy functions, with linear interpolation for offgrid capital values.

6.2 Asset Prices

Risk free rate. At each date t, the household's Euler equation pins down the (gross) quarterly risk free rate:

$$R_t^f = \frac{1}{\mathbb{E}_t[m_{t+1}]}, \qquad m_{t+1} = \beta\left(\frac{c_{t+1}}{c_t}\right)^{-\gamma}, \tag{11}$$

where the expectation is taken over all nine next period TFP states, weighted by $\mathbf{P}[z_t, \cdot]$.

Equity return. The realized gross quarterly return on the capital stock is the marginal product of capital plus undepreciated capital:

$$R_{t+1}^e = \alpha\, Z_{t+1}\, k_{t+1}^{\alpha-1} + (1-\delta). \tag{12}$$

Annualization. Both returns are annualized as $\bar{R}^{\text{ann}} = (\bar{R}^q)^4 - 1$, where $\bar{R}^q$ is the sample mean of the gross quarterly rate. The equity premium is $\mathbb{E}[R^e] - \mathbb{E}[R^f]$.

6.3 Quantitative Results

Table 7 is the central result of the project. It reports, for each γ, the annualized risk free rate, equity return, equity premium, and three moments of log consumption growth.

Table 7: Asset Pricing and Consumption Moments by Risk Aversion γ
(Annualized rates in %; simulation of N = 10,000 quarters)

γ	$\mathbb{E}[R^f]$ (%)	$\mathbb{E}[R^e]$ (%)	$\mathbb{E}[R^e - R^f]$ (%)	$\mathbb{E}[\Delta \log c]$	$\sigma(\Delta \log c)$	$\rho_1(\Delta \log c)$
0.5	4.096	4.110	0.014	-8.30×10^{-7}	0.0102	-0.488
1.5	4.053	4.109	0.055	-8.55×10^{-7}	0.0100	-0.489
3.0	4.055	4.121	0.066	-9.55×10^{-7}	0.0099	-0.491
5.0	4.086	4.124	0.038	-1.05×10^{-6}	0.0098	-0.494
10.0	4.148	4.144	-0.004	-1.25×10^{-6}	0.0097	-0.495

$\Delta \log c_t = \log c_t - \log c_{t-1}$; ρ_1 = first order autocorrelation of consumption growth; σ = standard deviation. Risk free rate and equity return annualized via $(\bar{R}^q)^4 - 1$. Burn in of 1,000 quarters discarded; `np.random.seed(323)`.

6.4 Discussion

An interesting finding in Table 7 is that the model generates an extremely negligible equity premium, which even turns *negative* at a high risk aversion of $\gamma = 10.0$. This is a manifestation of the Mehra and Prescott (1985) equity premium puzzle: a standard RBC model with CRRA preferences cannot replicate the historical U.S. equity premium of approximately 6% per year.
In equilibrium, the risk free rate equals the inverse of the expected stochastic discount factor (SDF). The equity return, pinned down by the marginal product of capital, does not co move sufficiently with the SDF to command a large, positive risk premium. Formally, the equity premium satisfies:

$$\mathbb{E}[R^e - R^f] \; = \; -\,(m_{t+1},\, R^e_{t+1})\, R^f, \tag{13}$$

and the covariance (m, R^e) is insufficiently negative in a calibrated RBC model because consumption growth is too smooth relative to asset returns.

As γ rises from 0.5 to 10.0, $\mathbb{E}[R^f]$ fluctuates moderately but stays high, between 4.05% and 4.15%. This is the *risk free rate puzzle* (Weil 1989): attempting to generate a high equity premium with higher γ simultaneously locks the risk free rate far above the historical average of approximately 1%.

Consumption Growth Moments

- **Mean.** $\mathbb{E}[\Delta \log c]$ is virtually zero (on the order of -10^{-7} to -10^{-6}) because the model is solved without a trend; the TFP process has a positive mean μ_g but the simulation horizon is finite and the initial capital is near the stochastic steady state.
- **Volatility.** $\sigma(\Delta \log c) \approx 1\%$ quarterly, corresponding to about 2% annualized, in line with post war U.S. data but far below aggregate stock return volatility ($\approx 17\%$), which is the core source of the puzzle.
- **Autocorrelation.** $\rho_1(\Delta \log c) \approx -0.49$ across all γ; consumption growth exhibits strong mean reversion. This is partly a grid induced artifact: the discrete capital grid introduces small oscillations in the interpolated policy function that show up as negative serial correlation.